Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/27/2006
PI Name:	Ma, Nancy Ph.D.		
Project Title:	Models of Mass Transport during Microgravity Crystal Growth of Alloyed Semiconductors in a Magnetic Field		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Raleigh	**State:**	NC
Zip Code:	27695-7919	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	05/01/2001	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Charles Rosenblatt		
COI Name (Institution):			
Grant/Contract No.:	NAG8-1817		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This grant involves two major tasks. The first involves the development of models which will treat the convection driven by compositional variations in the terrestrial vertical Bridgman process with an axial magnetic field in order to predict the composition of the entire alloyed crystal. These models would be validated by results from experiments which have been conducted at NASA MSFC. The second task will involve extending the terrestrial models to treat magnetically-damped compositionally-driven buoyant convection during Bridgman growth in space. The objective of the proposed research is to develop models which investigate the effects of the microgravity environment on segregation in alloyed semiconductor crystals. The significance of the results of this research is two-fold. First, these investigations would contribute to the fundamental physical understanding of the effects of terrestrial and microgravity as well as magnetic fields on compositionally-driven buoyant convection. Second, these models would guide process optimization for the production of alloyed semiconductor crystals such as GeSi, HgCdTe and PbTnTe, which are very important materials which serve as bases for optoelectronic devices.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A number of research projects were conducted under the support of NASA Grant NAG8-1817. These projects include (1) segregation in silicon-doped germanium semiconductor crystals grown by the Bridgman process with steady magnetic fields, (2) melt motion during floating-zone growth of high-purity silicon semiconductor crystals with rotating magnetic fields, and (3) melt motion during liquid-encapsulated Czochralski growth of compound semiconductor crystals with steady magnetic fields.
Bibliography Type:	Description: (Last Updated: 03/27/2006)
Articles in Peer-reviewed Journals	Wang X, Ma N. "Strong magnetic field asymptotic model for binary alloyed semiconductor crystal growth." AIAA Journal of Thermophysics and Heat Transfer. 2004 Oct-Dec;18(4):476-80. , Oct-2004